High Voltage Test Lab

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Test Facilities Provided in High Voltage Lab

- Transformers
- Lighting arrester
- Isolator and C.B
- Different type of insulators
- Cables
- Capacitors
- Line hardware and accessories
- Other equipment like reactors

Activities and Studies in High Voltage Lab

- Main is testing on equipment
- Other than it is research
- So research include

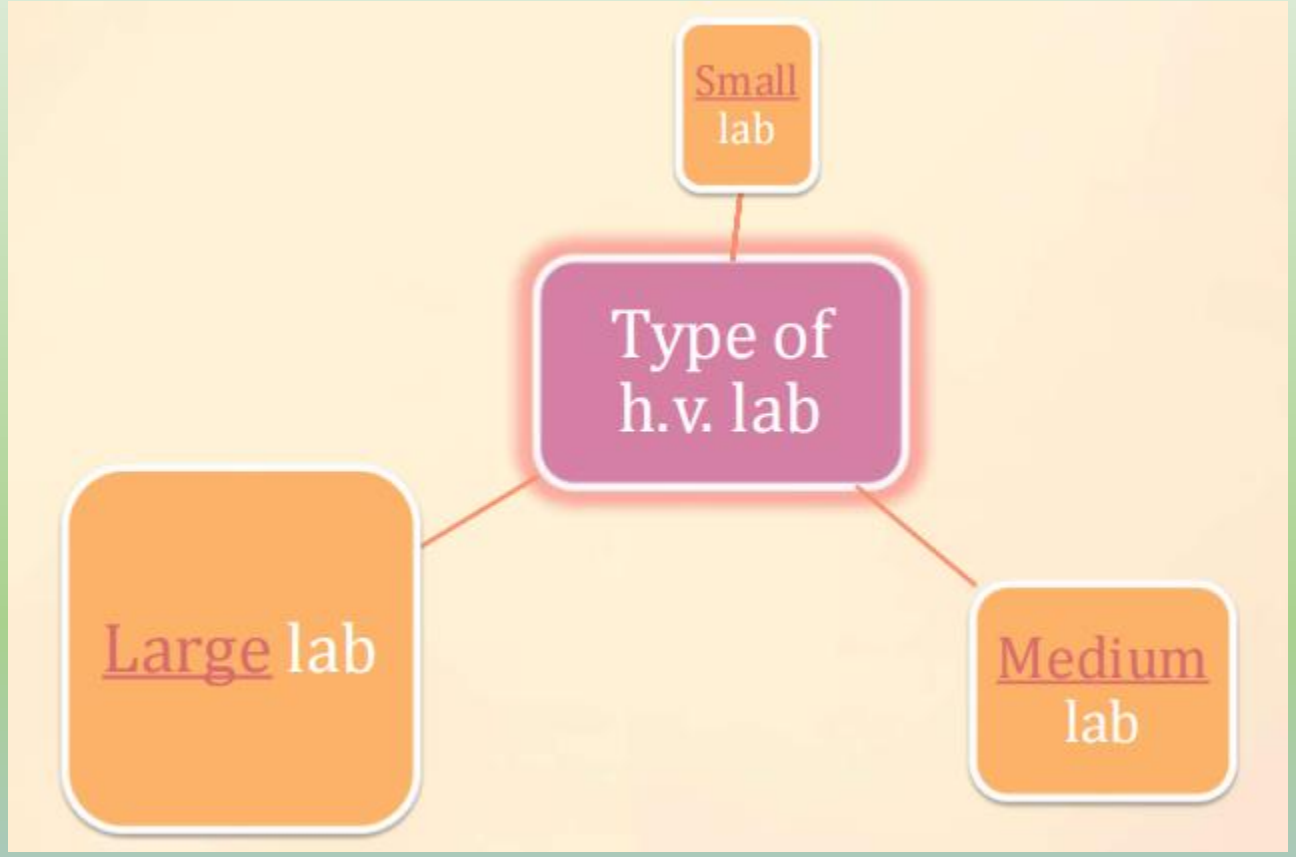
i. Breakdown strength of different material like solid ,liquid and gas.

ii. Electrical interference of transmission line

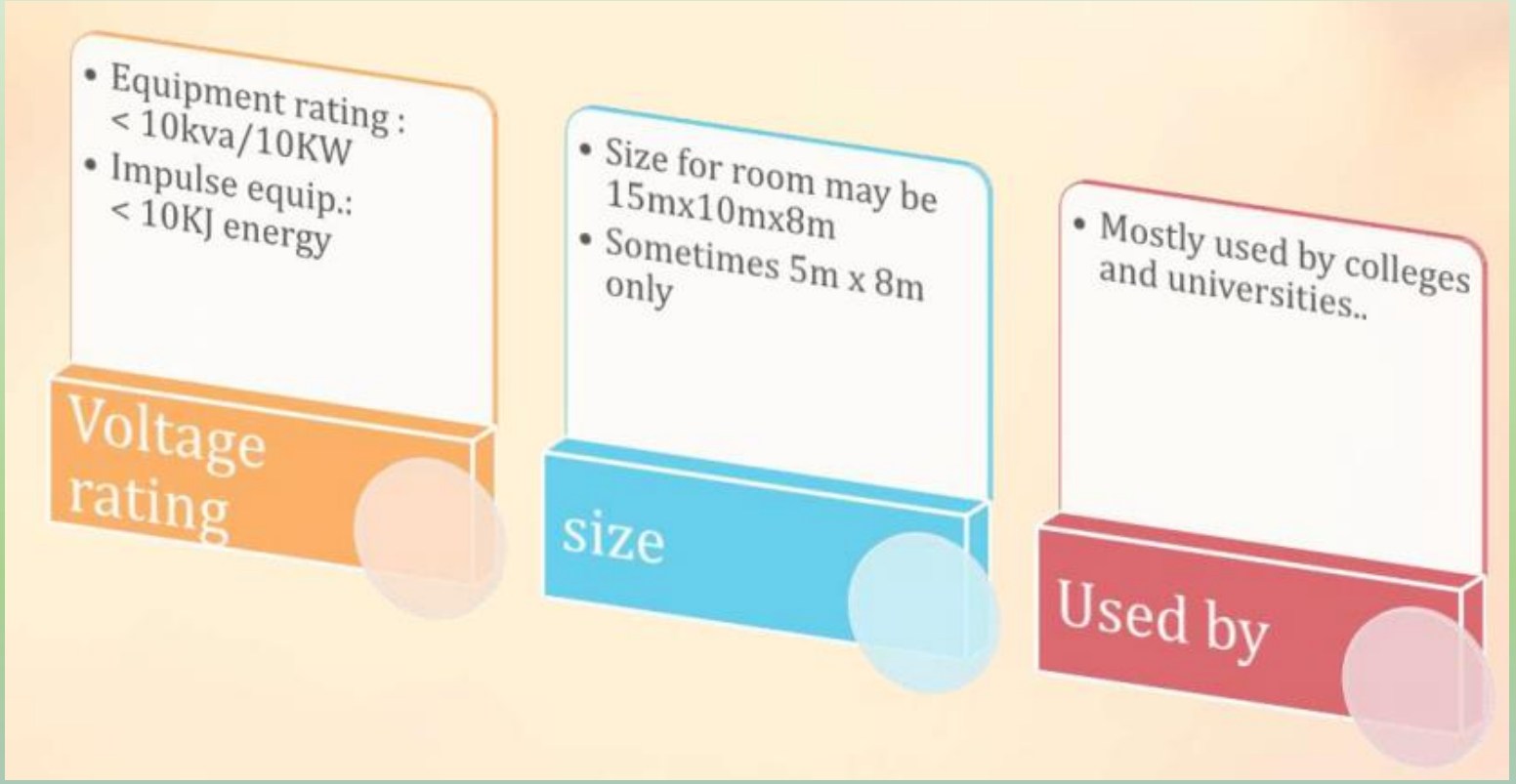
iii. High voltage power system's insulation

iv. High current phenomena such as electrics arc and plasma physics

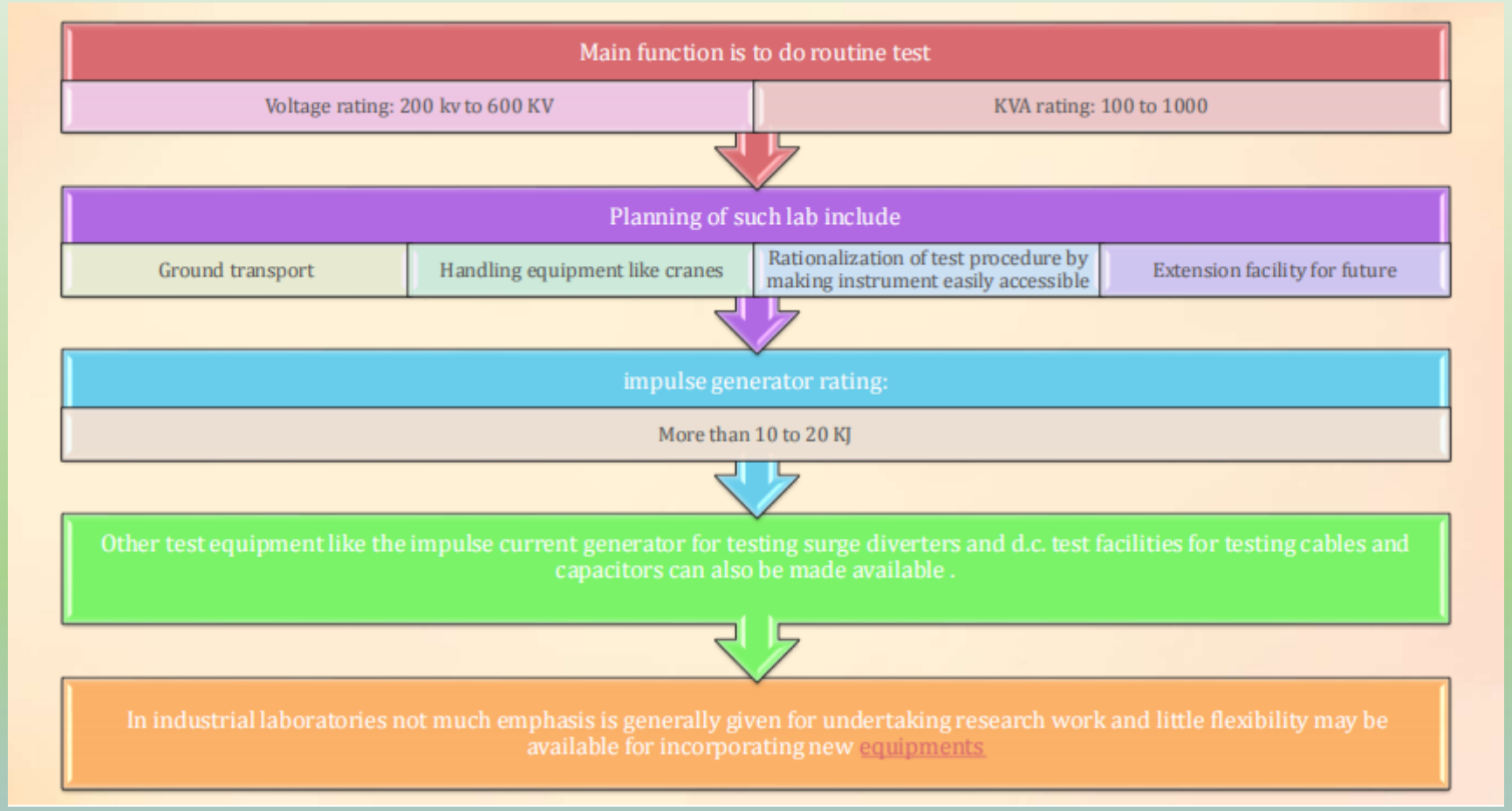
High Voltage Lab Classifications



Small High Voltage Lab



Medium High Voltage Lab



Large High Voltage Lab



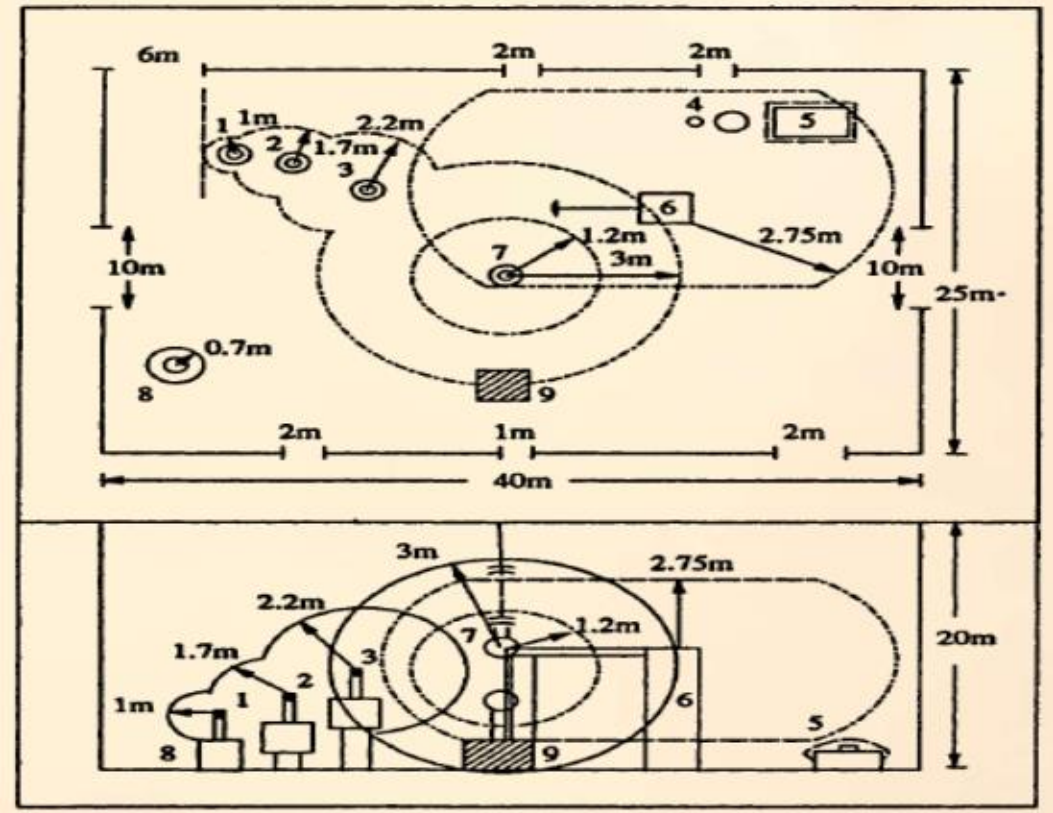
Large High Voltage Lab Components



Large High Voltage Lab



High Voltage Lab Layout

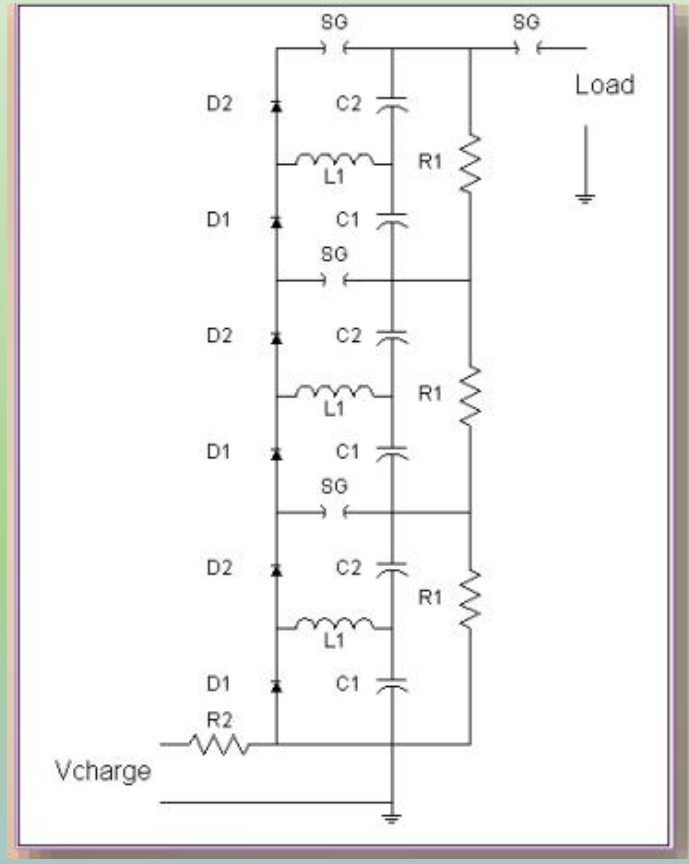


1,2,3	— Cascade transformer set	: 1 MV
4	— d.c. charging unit	: 200 kV
5	— Impulse current generator	: 200 kA
6	— Impulse voltage generator	: 3 MV
7	— Sphere gap	: 2 metres diameter
8	— d.c. test set	: 300 kV
9	— Control room Faraday cage	: 5 × 3.5 × 3 m^3

Control Panel of High Voltage Testing Instrument



Impulse Generator





Impulse Tests on Transformers





Impulse Voltage Generator 600 kV ,
18 kJ

Complete High Voltage Test Systems



Automatic HV and PD measuring systems for testing of bushings for SF_6 switchgears

Complete High Voltage Test Systems…







Specific test systems for different applications, like



- surge arresters
- switchgears

